

Paul Lintzen

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(831) 246-0427 | Denver & Golden, CO



EDUCATION

Colorado School of Mines, Golden CO

May 2027

Master of Science Non-Thesis in Data Science

GPA: 4.0

- Completing B.S. and M.S. in 4 years through intensive coursework and planning

Colorado School of Mines, Golden CO

Aug 2026

Bachelor of Science in Computer Science with a Data Science Focus

GPA: 4.0

- Tau Beta Pi Engineering Honor Society alumni; Summa Cum Laude
- 4-year President's Scholarship; Dean's List all semesters (2023–2026)

SKILLS

Technologies: Git/GitHub, Pandas, PyTorch, TensorFlow, APIs, CLIs (Windows CMD, Linux Bash), VSCode, RStudio, Excel, Windows OS, UML modeling, Agile Methodology, SolidWorks (CSWA certified), Word, PowerPoint

Languages: C++, C, Java, Python, SQL, R, RISC-V Assembly, LaTeX, OCaml and Bash

Communication: Technical writing, public speaking; multilingual – English/German/basic Spanish

PROJECTS

DNS Threat Detection, Colorado School of Mines

Apr - May 2026

- Built a two-stage DNS threat detection ensemble combining an XGBoost domain classifier ($F1=0.98$, $AUC-ROC=0.998$) with an Isolation Forest behavioral anomaly detector trained on 650k real-world Pi-hole queries.
- Engineered domain string and cyclical temporal features across both models, combining outputs via normalized ensemble scoring to surface threats across orthogonal feature spaces.

Linux Server Administration & Home Lab Infrastructure, Personal

Jan 2026 - Present

- Deployed Raspberry Pi and secondary Linux server for Pi-hole network-wide ad blocking, DHCP management, and multiplayer game hosting; set up firewall, port forwarding, and dynamic DNS configuration.
- Self-hosted WireGuard VPN with SSH key authentication for encrypted remote access and DNS filtering, enabling secure connectivity and ad blocking on the go.

AI Food Classifier & Calorie Predictor, Colorado School of Mines

Oct - Dec 2025

- Trained a CNN in TensorFlow on a 101k-image, 101-class food dataset, achieving 60% validation accuracy, and integrated nutrition APIs to estimate calorie information per class.
- Built Flask web app enabling real-time image upload, classification, calorie estimates, and confidence scores.

WORK EXPERIENCE

Quality Assurance Intern, Quantum Rings

July 2026 - Present

- Built a quantum circuit benchmarking pipeline, compiling and benchmarking a 15,000-run dataset in-house to measure performance across hardware and backend settings and compare SDK versions.
- Trained an XGBoost runtime prediction model to R^2 of 0.93, then refined it on a curated dataset of generated and hand-picked circuits to reach 80% bucket accuracy, tuning to minimize underprediction and avoid underpricing hosted quantum simulation jobs.

Graduate Machine Learning Researcher, Dr. Lowe's Research Group

May 2026 - Present

- Collaborate with a multi-disciplinary team applying deep learning methods to microscopy-based metal sheet failure analysis for medical device applications.
- Architect and train a hierarchical classifier on DINOv3 and ConvNeXt embeddings, using Optuna-tuned hyperparameters to reach 75% failure detection accuracy on unseen sheets.

Teacher's Assistant, CSCI406 Algorithms

Jan 2026 - Present

- Support ~130 students per semester through a notoriously demanding course, hosting office hours multiple times per week and conducting in-person grading interviews to verify genuine understanding of project work.